

IEEE Backend API Reference Documentation

Comprehensive API documentation for the **IEEE GBPIET Student Branch Backend**.

Summary of APIs

1. Global Metrics

- **Total APIs Count:** 30
- **Total Modules:** 7
- **Protected APIs (Require JWT Auth):** 17
- **Public APIs:** 13

2. Module-wise Breakdown

Module Name	Base Route	Total Endpoints	Public	Protected (Admin)
Health Check	/health	1	1	0
Authentication & OTP	/api/v1/auth	5	5	0
Certificates	/api/v1/certificate	5	1	4
Department Posts	/api/v1/department	5	2	3
Upcoming Events	/api/v1/upcomingevents	5	2	3
Support & Tickets	/api/v1/support	5	1	4
Dashboard Analytics	/api/v1/dashboard	4	0	4
Total		30	13	17

Complete Master API Table

#	Module	Method	Endpoint	Auth Required	Content-Type	Summary Description
1	Health Check	GET	/health	No	None	Server health status check
2	Auth	POST	/api/v1/auth/login	No (Rate Limited)	application/json	Admin login & JWT token generation
3	Auth	POST	/api/v1/auth/logout	No	None	Clears authentication session cookie
4	Auth	POST	/api/v1/auth/resetPassword/otp/generateOTP	No	application/json	Generates 6-digit OTP & emails admin
5	Auth	POST	/api/v1/auth/resetPassword/otp/verifyOtp	No	application/json	Verifies OTP & returns reset token

#	Module	Method	Endpoint	Auth Required	Content-Type	Summary Description
6	Auth	PATCH	/api/v1/auth/resetPassword	No (Reset Token)	application/json	Updates admin password using reset token
7	Certificate	POST	/api/v1/certificate/applynow	No	application/json	Participant applies for event certificate
8	Certificate	GET	/api/v1/certificate/all	Yes (Bearer)	None	Admin retrieves all certificate applications
9	Certificate	PATCH	/api/v1/certificate/approved/:id	Yes (Bearer)	None	Approves certificate, renders PDF & emails
10	Certificate	PATCH	/api/v1/certificate/rejected/:id	Yes (Bearer)	None	Rejects a certificate application
11	Certificate	POST	/api/v1/certificate/adminApply	Yes (Bearer)	application/json	Admin directly issues, renders PDF & emails
12	Department	POST	/api/v1/department/create	Yes (Bearer)	multipart/form-data	Creates department activity post + image upload
13	Department	GET	/api/v1/department/all	No	None	Retrieves department posts (optional ?dep=)
14	Department	GET	/api/v1/department/post/:id	No	None	Retrieves single department post by postId
15	Department	PATCH	/api/v1/department/edit/:id	Yes (Bearer)	multipart/form-data	Updates department post & replaces old image
16	Department	DELETE	/api/v1/department/delete/:id	Yes (Bearer)	None	Deletes department post & removes Cloudinary image

#	Module	Method	Endpoint	Auth Required	Content-Type	Summary Description
17	Upcoming Events	POST	/api/v1/upcomingevents/create	Yes (Bearer)	multipart/form-data	Creates upcoming event + banner upload
18	Upcoming Events	GET	/api/v1/upcomingevents/all	No	None	Retrieves all upcoming events
19	Upcoming Events	GET	/api/v1/upcomingevents/post/:id	No	None	Retrieves single event details by postId
20	Upcoming Events	PATCH	/api/v1/upcomingevents/edit/:id	Yes (Bearer)	multipart/form-data	Updates event details & replaces old banner
21	Upcoming Events	DELETE	/api/v1/upcomingevents/delete/:id	Yes (Bearer)	None	Deletes event & removes Cloudinary image
22	Support	POST	/api/v1/support/sendmsg	No	application/json	Public submits support inquiry / contact message
23	Support	GET	/api/v1/support/allticket	Yes (Bearer)	None	Admin retrieves all support tickets
24	Support	GET	/api/v1/support/viewTicket/:id	Yes (Bearer)	None	Admin views single support ticket details
25	Support	PATCH	/api/v1/support/closeTicket/:id	Yes (Bearer)	None	Admin marks ticket status as "solved"
26	Support	PATCH	/api/v1/support/Reject/:id	Yes (Bearer)	None	Admin marks ticket status as "rejected"
27	Dashboard	GET	/api/v1/dashboard/departmentposts/getall	Yes (Bearer)	None	Department post counts grouped by branch
28	Dashboard	GET	/api/v1/dashboard/events/getall	Yes (Bearer)	None	Top 3 upcoming events for dashboard widget

#	Module	Method	Endpoint	Auth Required	Content-Type	Summary Description
29	Dashboard	GET	/api/v1/dashboard/contactus/getall	Yes (Bearer)	None	Ticket counts grouped by solvedStatus
30	Dashboard	GET	/api/v1/dashboard/certificates/getall	Yes (Bearer)	None	Certificate counts grouped by status

Authentication Details

For all endpoints marked **Auth Required: Yes**:

- Provide the token in the **Authorization** header:

```
Authorization: Bearer <your_jwt_token>
```

- Alternatively, you may provide:
 - Header: **x-access-token**: <your_jwt_token>
 - Cookie: **Token**=<your_jwt_token>

If unauthorized or token is expired:

```
{
  "success": false,
  "message": "Access denied. No token provided."
}
```

Or:

```
{
  "success": false,
  "message": "Invalid or expired token."
}
```

Detailed Endpoint Specifications

1. Health Check Module

GET /health

Verifies server uptime and connectivity.

- **Authentication**: None
- **Headers**: None
- **Response (200 OK)**:

```
{
  "success": true,
  "message": "healthy"
}
```

2. Authentication Module (/api/v1/auth)

POST /api/v1/auth/login

Authenticates administrator account (ieee@gbpiet.ac.in). Protected with rate limiting (max 10 attempts per 15 minutes).

- **Authentication:** None
- **Headers:** Content-Type: application/json
- **Request Body:**

```
{
  "email": "ieee@gbpiet.ac.in",
  "password": "AdminPassword123"
}
```

- **Response (200 OK):**

```
{
  "msg": "Login Successfull",
  "success": true,
  "user": {
    "id": "6741ab82cd98ef1234567890",
    "email": "ieee@gbpiet.ac.in"
  },
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

- **Response (401 Unauthorized):**

```
{
  "success": false,
  "message": "Incorrect password"
}
```

- **Response (429 Too Many Requests):**

```
{
  "success": false,
  "message": "Too many login attempts. Please try again later."
}
```

POST /api/v1/auth/logout

Clears administrative authentication cookie.

- **Authentication:** None
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Logout successful"
}
```

POST /api/v1/auth/resetPassword/otp/generateOTP

Sends a cryptographically generated 6-digit OTP to the administrator email address.

- **Authentication:** None
- **Headers:** Content-Type: application/json
- **Request Body:**

```
{
  "email": "ieee@gbpiet.ac.in"
}
```

- **Response (200 OK):**

```
{
  "success": true,
  "message": "Password reset OTP sent successfully"
}
```

- **Response (400 Bad Request):**

```
{
  "success": false,
  "message": "Unauthorized email address"
}
```

POST /api/v1/auth/resetPassword/otp/verifyOtp

Validates the 6-digit OTP and generates a temporary `resetToken` (valid for 10 minutes).

- **Authentication:** None
- **Headers:** Content-Type: application/json
- **Request Body:**

```
{
  "email": "ieee@gbpiet.ac.in",
  "otp": "482910"
}
```

- **Response (200 OK):**

```
{
  "success": true,
  "message": "OTP verified successfully",
  "resetToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

- **Response (400 Bad Request):**

```
{
  "success": false,
  "message": "Invalid OTP. 2 attempt(s) remaining."
}
```

PATCH /api/v1/auth/resetPassword

Resets the administrator password using the verified `resetToken`.

- **Authentication:** None (`resetToken` provided in body or header)
- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "newPassword": "NewStrongPassword@2026",
  "resetToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

- **Response (200 OK):**

```
{
  "success": true,
  "message": "Password reset successfully"
}
```

- **Response (400 Bad Request):**

```
{
  "success": false,
  "message": "Invalid or expired reset token"
}
```

3. Certificate Module (/api/v1/certificate)

POST /api/v1/certificate/applynow

Allows students / participants to apply for an event certificate. Creates a record with `status: "pending"`.

- **Authentication:** None
- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "name": "Rohan Sharma",
}
```

```
{
  "email": "rohan.sharma@gmail.com",
  "branch": "CSE",
  "event": "National Hackathon 2026",
  "date": "2026-03-20",
  "position": "1st"
}
```

(Note: *position* is optional. If provided, must be "1st", "2nd", or "3rd").

- **Response (201 Created):**

```
{
  "success": true,
  "message": "Application submitted successfully. Awaiting admin approval.",
  "data": {
    "_id": "6732ef12bc093412a8765432",
    "name": "Rohan Sharma",
    "email": "rohan.sharma@gmail.com",
    "branch": "CSE",
    "eventName": "National Hackathon 2026",
    "date": "2026-03-20",
    "position": "1st",
    "certificateId": "IEEE-CERT-9A82D10E",
    "status": "pending",
    "createdAt": "2026-03-25T14:32:00.000Z",
    "updatedAt": "2026-03-25T14:32:00.000Z"
  }
}
```

- **Response (400 Bad Request):**

```
{
  "success": false,
  "message": "All fields (name, email, event, branch, date) are required"
}
```

GET /api/v1/certificate/all

Fetches all certificate applications across all statuses (*pending*, *approved*, *rejected*).

- **Authentication: Required** (*verifyToken*)
- **Headers:** *Authorization: Bearer <token>*
- **Response (200 OK):**

```
{
  "success": true,
  "count": 2,
  "data": [
    {
      "_id": "6732ef12bc093412a8765432",
      "name": "Rohan Sharma",
      "email": "rohan.sharma@gmail.com",
      "branch": "CSE",
      "eventName": "National Hackathon 2026",
      "date": "2026-03-20",
      "position": "1st",
      "certificateId": "IEEE-CERT-9A82D10E",
      "status": "pending",
      "createdAt": "2026-03-25T14:32:00.000Z"
    }
  ]
}
```

```
}  
]  
}
```

PATCH /api/v1/certificate/approved/:id

Approves a pending certificate application. Triggers **Puppeteer** to generate the PDF certificate and **Resend** to email the PDF attachment directly to the applicant.

- **Authentication: Required** (**verifyToken**)
- **Headers:** **Authorization:** Bearer <token>
- **URL Parameters:** **:id** (MongoDB **_id** of certificate)
- **Response (200 OK):**

```
{  
  "success": true,  
  "message": "Certificate approved and emailed to the applicant successfully",  
  "data": {  
    "_id": "6732ef12bc093412a8765432",  
    "name": "Rohan Sharma",  
    "email": "rohan.sharma@gmail.com",  
    "certificateId": "IEEE-CERT-9A82D10E",  
    "status": "approved",  
    "updatedAt": "2026-03-25T14:35:12.000Z"  
  },  
  "email": {  
    "messageId": "re_abc123xyz"  
  }  
}
```

- **Response (400 Bad Request):**

```
{  
  "success": false,  
  "message": "Certificate is already \"approved\" and cannot be approved"  
}
```

- **Response (404 Not Found):**

```
{  
  "success": false,  
  "message": "Certificate not found"  
}
```

- **Response (502 Bad Gateway):**

```
{  
  "success": false,  
  "message": "Certificate approved but email delivery failed"  
}
```

PATCH /api/v1/certificate/rejected/:id

Rejects a pending certificate application.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (MongoDB `_id` of certificate)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Certificate rejected successfully",
  "data": {
    "_id": "6732ef12bc093412a8765432",
    "status": "rejected",
    "updatedAt": "2026-03-25T14:36:00.000Z"
  }
}
```

POST /api/v1/certificate/adminApply

Directly generates, approves, and emails a certificate to a recipient without waiting for an approval step.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`, `Content-Type: application/json`
- **Request Body:**

```
{
  "name": "Priya Rawat",
  "email": "priya.rawat@gmail.com",
  "branch": "AIML",
  "event": "Generative AI Summit",
  "date": "2026-03-22",
  "position": "2nd"
}
```

- **Response (201 Created):**

```
{
  "success": true,
  "message": "Certificate issued and emailed successfully",
  "data": {
    "_id": "6732f012bc093412a8765999",
    "name": "Priya Rawat",
    "email": "priya.rawat@gmail.com",
    "certificateId": "IEEE-CERT-8B12C33F",
    "status": "approved"
  },
  "email": {
    "messageId": "re_xyz789def"
  }
}
```

4. Department Module (/api/v1/department)

POST /api/v1/department/create

Creates a departmental activity report/post with optional Cloudinary image upload.

- **Authentication: Required** (`verifyToken`)
- **Headers:**

- Authorization: Bearer <token>
- Content-Type: multipart/form-data
- **Query Parameters:** ?dep=CSE (optional, can also be provided in *branch* form field)
- **FormData Fields:**
 - title (string, required): Headline of the post
 - category (string, required): E.g., "Workshop", "Guest Lecture", "Coding Contest"
 - branch (string, required if ?dep= not set): "CSE", "AIML", "EE", "ECE", or "BT"
 - date (string, required): "2026-03-25"
 - time (string, optional): "10:00 AM - 01:00 PM"
 - venue (string, required): "Computer Lab 3, GBPIET"
 - organizedBy (string, required): "IEEE Computer Society Chapter"
 - reportAuthor (string, optional): "Dr. V. K. Sharma"
 - overview (string, required): Executive summary paragraph
 - description (string, required): Detailed narrative report
 - keyDiscussion (string or JSON array): ["Python Basics", "Neural Networks", "Hands-on PyTorch"]
 - studentsPresent (string or JSON array): ["Rahul Verma", "Sneha Bist", "Ankit Negi"]
 - image (file, optional): Image file (jpg, jpeg, png, webp, max 5MB)
- **Response (201 Created):**

```
{
  "success": true,
  "message": "Post created successfully",
  "post": {
    "_id": "67330012bc093412a8766111",
    "id": "CSE-20260325-1042",
    "postId": "CSE-20260325-1042",
    "title": "Hands-on Deep Learning Workshop",
    "category": "Workshop",
    "branch": "CSE",
    "date": "2026-03-25",
    "time": "10:00 AM - 01:00 PM",
    "venue": "Computer Lab 3, GBPIET",
    "organizedBy": "IEEE Computer Society Chapter",
    "reportAuthor": "Dr. V. K. Sharma",
    "overview": "A comprehensive deep learning workshop covering fundamentals to CNNs.",
    "description": "The workshop was attended by 65 participants from across Uttarakhand...",
    "keyDiscussion": ["Python Basics", "Neural Networks", "Hands-on PyTorch"],
    "studentsPresent": ["Rahul Verma", "Sneha Bist", "Ankit Negi"],
    "image": {
      "url": "https://res.cloudinary.com/ieee-gbpiet/image/upload/v1727280000/ieee-department-posts/abc123xyz.png",
      "publicId": "ieee-department-posts/abc123xyz"
    },
    "createdAt": "2026-03-25T15:00:00.000Z",
    "updatedAt": "2026-03-25T15:00:00.000Z"
  }
}
```

- **Response (400 Bad Request):**

```
{
  "success": false,
  "message": "File size exceeds the 5MB limit."
}
```

GET /api/v1/department/all

Fetches published department posts. Supports optional filtering by academic branch.

- **Authentication:** None
- **Query Parameters:**
 - `?dep=CSE` (optional, filter by "CSE", "AIML", "EE", "ECE", or "BT")
- **Response (200 OK):**

```
{
  "success": true,
  "count": 1,
  "posts": [
    {
      "_id": "67330012bc093412a8766111",
      "postId": "CSE-20260325-1042",
      "title": "Hands-on Deep Learning Workshop",
      "category": "Workshop",
      "branch": "CSE",
      "date": "2026-03-25",
      "image": {
        "url": "https://res.cloudinary.com/.../abc123xyz.png",
        "publicId": "ieee-department-posts/abc123xyz"
      },
      "createdAt": "2026-03-25T15:00:00.000Z"
    }
  ]
}
```

GET /api/v1/department/post/:id

Retrieves single department post by unique `postId`.

- **Authentication:** None
- **URL Parameters:** `:id` (e.g. `CSE-20260325-1042`)
- **Response (200 OK):**

```
{
  "success": true,
  "post": {
    "_id": "67330012bc093412a8766111",
    "postId": "CSE-20260325-1042",
    "title": "Hands-on Deep Learning Workshop",
    "branch": "CSE",
    "overview": "...",
    "description": "...",
    "image": {
      "url": "https://res.cloudinary.com/...",
      "publicId": "..."
    }
  }
}
```

- **Response (500 Internal Server Error):**

```
{
  "success": false,
  "message": "Failed to fetch post"
}
```

PATCH /api/v1/department/edit/:id

Updates existing department post. If a new image is uploaded, the existing Cloudinary asset is automatically destroyed.

- **Authentication: Required** (`verifyToken`)
- **Headers:**
 - `Authorization: Bearer <token>`
 - `Content-Type: multipart/form-data` or `application/json`
- **URL Parameters:** `:id` (`postId`)
- **FormData / JSON Body:** Any subset of fields to update, plus optional `image` file
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Post updated successfully",
  "post": {
    "postId": "CSE-20260325-1042",
    "title": "Hands-on Deep Learning Workshop (Updated)",
    "updatedAt": "2026-03-25T15:30:00.000Z"
  }
}
```

DELETE /api/v1/department/delete/:id

Deletes a department post by `postId`. Deletes the associated Cloudinary asset and MongoDB document.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (`postId`)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Post deleted successfully"
}
```

5. Upcoming Events Module (/api/v1/upcomingevents)

POST /api/v1/upcomingevents/create

Publishes a new upcoming event with optional banner image.

- **Authentication: Required** (`verifyToken`)
- **Headers:**
 - `Authorization: Bearer <token>`
 - `Content-Type: multipart/form-data`
- **FormData Fields:**
 - `eventName` (*string, required*): "Robotics & Embedded Systems Bootcamp"
 - `title` (*string, required*): "Learn Arduino, ESP32, and ROS2"
 - `date` (*date string, required*): "2026-04-15"
 - `lastDate` (*date string, required*): "2026-04-10"
 - `overview` (*string, required*): "3-day intensive hardware bootcamp..."
 - `image` (*file, optional*): Banner image (max 5MB)
- **Response (201 Created):**

```
{
  "success": true,
  "message": "Event created successfully",
}
```

```
"post": {
  "_id": "67331112bc093412a8767222",
  "postId": "EVT-93821",
  "eventName": "Robotics & Embedded Systems Bootcamp",
  "title": "Learn Arduino, ESP32, and ROS2",
  "date": "2026-04-15T00:00:00.000Z",
  "lastDate": "2026-04-10T00:00:00.000Z",
  "overview": "3-day intensive hardware bootcamp...",
  "image": {
    "url": "https://res.cloudinary.com/ieee-gbpiet/image/upload/v1727280000/ieee-upcoming-events/banner1.png",
    "publicId": "ieee-upcoming-events/banner1"
  },
  "createdAt": "2026-03-25T16:00:00.000Z"
}
```

GET /api/v1/upcomingevents/all

Fetches all published upcoming events sorted by latest.

- **Authentication:** None
- **Response (200 OK):**

```
{
  "success": true,
  "count": 1,
  "posts": [
    {
      "postId": "EVT-93821",
      "eventName": "Robotics & Embedded Systems Bootcamp",
      "date": "2026-04-15T00:00:00.000Z",
      "lastDate": "2026-04-10T00:00:00.000Z",
      "image": {
        "url": "https://res.cloudinary.com/..."
      }
    }
  ]
}
```

GET /api/v1/upcomingevents/post/:id

Retrieves single upcoming event by `postId`.

- **Authentication:** None
- **URL Parameters:** `:id` (e.g. `EVT-93821`)
- **Response (200 OK):**

```
{
  "success": true,
  "post": {
    "postId": "EVT-93821",
    "eventName": "Robotics & Embedded Systems Bootcamp",
    "overview": "..."
  }
}
```

PATCH /api/v1/upcomingevents/edit/:id

Updates an upcoming event by `postId`. Automatically cleans up the previous Cloudinary banner if a new banner is uploaded.

- **Authentication: Required** (`verifyToken`)
- **Headers:**
 - `Authorization: Bearer <token>`
 - `Content-Type: multipart/form-data` or `application/json`
- **URL Parameters:** `:id` (`postId`)
- **FormData / JSON Body:** subset of fields (`eventName`, `title`, `date`, `lastDate`, `overview`, `image`)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Event updated successfully",
  "post": {
    "postId": "EVT-93821",
    "eventName": "Robotics & Embedded Systems Bootcamp (Extended)"
  }
}
```

DELETE /api/v1/upcomingevents/delete/:id

Deletes an event by `postId`. Deletes the associated Cloudinary banner asset and MongoDB record.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (`postId`)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Event deleted successfully"
}
```

6. Support Module (/api/v1/support)

POST /api/v1/support/sendmsg

Submits a public inquiry or support request. Generates an alphanumeric ticket identifier (e.g. `TKT12A9B8C7`).

- **Authentication:** None
- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "name": "Aman Joshi",
  "email": "aman.joshi@example.com",
  "subject": "Inquiry regarding IEEE Membership Benefits",
  "message": "Could you provide details on student membership fee subsidies?"
}
```

- **Response (201 Created):**

```
{
  "success": true,
  "message": "Message sent successfully",
  "ticket": {
    "_id": "67332212bc093412a8768333",
    "ticketId": "TKT12A9B8C7",
    "name": "Aman Joshi",
    "email": "aman.joshi@example.com",
    "subject": "Inquiry regarding IEEE Membership Benefits",
    "description": "Could you provide details on student membership fee subsidies?",
    "solvedStatus": "pending",
    "createdAt": "2026-03-25T16:30:00.000Z",
    "updatedAt": "2026-03-25T16:30:00.000Z"
  }
}
```

GET /api/v1/support/allticket

Retrieves all support tickets sorted in reverse chronological order.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **Response (200 OK):**

```
{
  "success": true,
  "count": 1,
  "tickets": [
    {
      "_id": "67332212bc093412a8768333",
      "ticketId": "TKT12A9B8C7",
      "name": "Aman Joshi",
      "email": "aman.joshi@example.com",
      "subject": "Inquiry regarding IEEE Membership Benefits",
      "solvedStatus": "pending",
      "createdAt": "2026-03-25T16:30:00.000Z"
    }
  ]
}
```

GET /api/v1/support/viewTicket/:id

Retrieves full details of a specific ticket by MongoDB `_id`.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (MongoDB ObjectID)
- **Response (200 OK):**

```
{
  "success": true,
  "ticket": {
    "_id": "67332212bc093412a8768333",
    "ticketId": "TKT12A9B8C7",
    "name": "Aman Joshi",
    "email": "aman.joshi@example.com",
    "subject": "Inquiry regarding IEEE Membership Benefits",
    "description": "Could you provide details on student membership fee subsidies?",
  }
}
```

```
"solvedStatus": "pending",
"createdAt": "2026-03-25T16:30:00.000Z"
}
}
```

PATCH /api/v1/support/closeTicket/:id

Marks a support ticket as resolved (`solvedStatus: "solved"`).

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (MongoDB ObjectID)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Ticket closed successfully",
  "ticket": {
    "_id": "67332212bc093412a8768333",
    "ticketId": "TKT12A9B8C7",
    "solvedStatus": "solved",
    "updatedAt": "2026-03-25T16:45:00.000Z"
  }
}
```

PATCH /api/v1/support/Reject/:id

Marks a support ticket as rejected (`solvedStatus: "rejected"`).

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **URL Parameters:** `:id` (MongoDB ObjectID)
- **Response (200 OK):**

```
{
  "success": true,
  "message": "Ticket rejected successfully",
  "ticket": {
    "_id": "67332212bc093412a8768333",
    "ticketId": "TKT12A9B8C7",
    "solvedStatus": "rejected",
    "updatedAt": "2026-03-25T16:45:00.000Z"
  }
}
```

7. Dashboard Module (/api/v1/dashboard)

All dashboard routes are administrative and protected by `verifyToken`.

GET /api/v1/dashboard/departmentposts/getall

Aggregates post counts across the 5 engineering branches (CSE, AIML, EE, ECE, BT).

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **Response (200 OK):**

```
{
  "success": true,
  "data": {
    "CSE": 14,
    "AIML": 9,
    "EE": 4,
    "ECE": 6,
    "BT": 2
  }
}
```

GET /api/v1/dashboard/events/getall

Retrieves top 3 recent upcoming events for dashboard quick view.

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **Response (200 OK):**

```
{
  "success": true,
  "data": [
    {
      "eventName": "Robotics & Embedded Systems Bootcamp",
      "lastDate": "2026-04-10T00:00:00.000Z"
    },
    {
      "eventName": "IEEE Day Hackathon 2026",
      "lastDate": "2026-04-20T00:00:00.000Z"
    }
  ]
}
```

GET /api/v1/dashboard/contactus/getall

Aggregates support inquiry statistics grouped by status (`pending`, `rejected`, `solved`).

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`
- **Response (200 OK):**

```
{
  "success": true,
  "data": {
    "pending": 5,
    "rejected": 2,
    "solved": 28
  }
}
```

GET /api/v1/dashboard/certificates/getall

Aggregates certificate metrics grouped by application status (`pending`, `approved`, `rejected`).

- **Authentication: Required** (`verifyToken`)
- **Headers:** `Authorization: Bearer <token>`

- **Response (200 OK):**

```
{
  "success": true,
  "data": {
    "pending": 12,
    "approved": 154,
    "rejected": 4
  }
}
```